

Same doc from last lab, **new section on UI Testing at the end!**

Business Rule Testing (*not testing CRUD here, see next section*)

EP TCIs

- **name**
- **gpa**

TCI	Parameter	Range of Inputs
EP1	gpa	0..4.0
EP2	name	1..255

EP Test Cases (I am not covering values outside of the 2 EPs, as they will be caught in BVA testing)

Test Case ID	TCI Covered	gpa	name	Exp.
TC1.1	EP1[2]	2.5	Jim Hopper	Accepted as valid

BV TCIs

- **name**
- **gpa**

TCI	Parameter	Boundary Value
BV1*	gpa	-1.0
BV2		0
BV3		1.0
BV4		2.0
BV5		3.99

BV6		4.0
BV7*		4.1
BV8*	name	NULL
BV9		1
BV10		2
BV11		254
BV12		255
BV13*		256

BV Test Cases (grouping test cases together that yield the same results to optimize, [] = same outcome)

Test Case ID	TCI Covered	gpa	name	Exp.
BVA1.1*	BV1	-1.0	Jim Hopper	Exception: gpa outside of range
BVA1.2	BV2[3][4][5][6]	0.0	Jim Hopper	Accepted as valid
BVA1.3*	BV8[13]	3.5	Some name over 255 ch. OR no name :)	Exception: name outside of range
BVA1.4	BV9[10][11][12]	4.0	J	Accepted as valid
BVA1.5*	BV6	4.1	Jim Hopper	Exception: gpa outside range

DT Testing (not retesting cases from EP/BVA)

Cause

- Course Full?

Effect

- False = Can add student (rule 1)
- True = Cannot add student (rule 2)

Test Case ID	TCI Covered	gpa	name	Course Full?	Exp
DT1.1	DT1	3.5	Jim Hopper	F	Can add student
DT1.2	DT2	3.5	Jim Hopper	T	Cannot add student

CRUD Process Testing (*illustrated as state transitions*)

Student States

- Student does not exist yet
- Create student
- Update student info
- Delete a student

Student Transitions (like a TCI)

API Command	Current State	Resulting State
POST (create student)	DNE	Student created
PUT (update student)	Student created	Student info updated
DELETE (delete student)	Student created/updated	Student deleted
GET (get student)	Student created/updated	Student info read

Student Transitions (Test Cases)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST1.1	Student DNE	POST <code>"/{id}"</code>	Created	Student created
ST1.2	Student created	GET <code>"/{id}"</code>	Created	Student info returned
ST1.3	Student created	PUT <code>"/{id}"</code>	Updated	Student info updated
ST1.4	Student created	DELETE <code>"/{id}"</code>	Deleted	Student removed
ST1.5*	Student DNE	PUT <code>"/{id}"</code>	Student DNE	Error
ST1.6*	Student DNE	GET <code>"/{id}"</code>	Student DNE	Error

Course Transitions (like a TCI)

- Course does not exist yet
- Create course
- Update course info
- Delete a course

API Command	Current State	Resulting State
POST (create course)	DNE	Course created
PUT (update course)	Course created	Course info updated
DELETE (delete course)	Course created/updated	Course deleted

GET (get all courses)	Course created/updated	Course info returned
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Course Transitions (Test Case)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST2.1	Course DNE	POST /courses/{id }	Created	Course created
ST2.2	Course(s) created	PUT /courses/{id }	Updated	Course info updated
ST2.3	Course(s) created	DELETE /courses/{id }	Removed	Course deleted
ST2.4	Course(s) created	GET /courses/{id }	Retrieved	Course(s) retrieved
ST2.44*	Course DNE	PUT /courses/{id }	Course DNE	Error
ST2.45*	Course DNE	GET /courses/{id }	Course DNE	Error

Student + Course State Transitions (like a TCI)

- Add student to an existing course with an empty roster
- Add student to an existing course with students
- Delete student from an existing course with only that student
- Delete student from an existing course with many students

API Command	Current State	Resulting State
POST (add student to a course)	Empty Course	Has Student(s)
POST (add student to a course)	Has student(s)	Full
DELETE (remove student from course)	Has student	Empty
DELETE (remove student from course)	Full	Has student(s)

Student + Course State Transitions (Test Cases)

Test Case ID	Current State	API Command	Resulting State	Exp.
ST2.5	Empty course	POST <code>/ {courseId} / students / {studentId}</code>	Has student(s)	Valid operation
ST2.6	Has student(s)	POST <code>/ {courseId} / students / {studentId}</code>	Full	Valid operation
ST2.7*	Full course	POST <code>/ {courseId} / students / {studentId}</code>	Full course	Invalid operation
ST2.8	Has student(s)	DELETE <code>/ {courseId} / students / {studentId}</code>	Empty/Has student(s)	Valid operation

UI Testing (Selenium)

User Story

As a user/registar, I want to manage students and courses in the web interface in order to maintain a class registration system.

Acceptance Criteria

Name	Criterion
SUI1	Create a valid student
SUI2	Create a student with GPA = 0.0
SUI3	Create a student with GPA = 4.0
SUI4*	Identify a user input error (invalid GPA)
SUI5*	Identify a user input error (name > 255 ch)
SUI6	Edit student
SUI7	Delete student
CUI1	Create a valid course
CUI2	Edit course
CUI3	Add student to course
CUI4	Remove student from course
CUI5*	Prevent adding student to full course
CUI6	Delete a course

Identify Components

Page	HTML Element	ID
Student List	input = text	new-student-name
	input = text	new-student-major
	input = text	new-student-gpa
	table	student-list-table
	button	delete-student-button
	button	edit-student-button
	input = text	edit-student-name
	input = text	edit-student-major
	input = text	edit-student-gpa
	button	edit-student-save-button
Course List	input = text	new-course-name
	input = text	new-course-instructor
	input = text	new-course-max-size
	input = text	new-course-room
	table	course-list-table
	button	edit-course-button
	button	delete-course-button
	select = dropdown/button	select-student
	button	add-student-button
	select = dropdown/button	remove-student-button

TCIs & Test Cases

ID	TCI	Inputs	Exp.
TC1	SUI1	Enter "Jim Hopper" Enter "Criminal Justice" Enter 3.5 Click create	Student added
TC2	SUI2	Enter "Jim Hopper" ... Enter 0.0 Click create	Student added
TC3	SUI3	Enter "Jim Lee" .. Enter 4.0 Click create	Student added
TC4*	SUI4	Enter GPA = -1.0	Student should not appear
TC5*	SUI5	Enter "name > 255" Click create	Second student should not appear
TC6	SUI6	Click edit Change name, major, or GPA Save	Updated student info appears
TC7	SUI7	Select student Click delete	Student removed
TC8	CUI1	Enter "Software Testing" name Enter "Dr. Baarsch" Enter 2 Enter 339	Course added
TC9	CUI2	Select course Click edit	Updated course info
TC10	CUI3	Select student Click add	Student added to roster

TC11	CUI4	Open course edit Select student Click delete	Student deleted from roster
TC12*	CUI5	Course size = 1 Add 2 students	Second student not added
TC11	CUI6	Select course Click delete	Course deleted